

Can We Steer Away Unfaithful Chain-of-Thought?

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Project Type

Type 1: Extending an existing paper.

Paper

Arcuschin, Janiak, Krzyzanowski, Rajamanoharan, Nanda & Conmy, *Chain-of-Thought Reasoning in the Wild Is Not Always Faithful*, ICML 2026 ([arXiv:2503.08679](https://arxiv.org/abs/2503.08679)).

What the paper shows

The paper presents models with pairs of logically opposite Yes/No questions—for instance, “Is X south of Y?” versus “Is Y south of X?”—and checks whether the model gives consistent answers. A faithful reasoner must answer oppositely; yet models sometimes give the same answer to both and produce different plausible-sounding justifications each time. The authors term this **Implicit Post-Hoc Rationalization (IPHR)** and report rates from 0.04 % to 13.49 % across 16 frontier models on 4,834 question pairs.

Crucially, a preliminary probing experiment (Appendix H) reveals that the Yes/No bias is already encoded in the model’s residual-stream activations *before* any chain-of-thought is generated. However, the authors perform no causal intervention and explicitly note that testing whether ablating the candidate template-bias direction reduces unfaithfulness is an open question.

Our question

Does the pre-reasoning bias direction identified by a linear probe causally contribute to the Yes/No answer bias underlying IPHR, and can activation ablation reduce IPHR without degrading accuracy?

Approach

We work with a small open model not evaluated in the original study, like Qwen3-4B-Instruct, running all experiments on Modal (A100; estimated budget \$70–120 of our \$250 allocation). The project proceeds in four stages.

Week 1 [Reproduction] - We run the paper’s IPHR evaluation pipeline on 500–1000 of their question pairs (10 rollouts each) to obtain a baseline IPHR rate for Qwen3-4B-Instruct.

Week 2 [Probing] - We cache residual-stream activations at the last prompt token across all layers and train a linear probe per layer to predict template-level Yes/No bias. The learned weight vector at the best-performing layer serves as our candidate bias direction.

Week 3 [Causal intervention] - Using a forward hook, we project out the bias direction during inference and re-run the IPHR evaluation. We include two controls: a random-direction ablation (to confirm the effect is specific to the bias direction) and an accuracy check on the same questions (to confirm the model is not simply degraded).

Week 4 [Analysis and write-up] - We consolidate results, prepare figures and the final report. As a secondary experiment, we run the IPHR evaluation on our risky-finance LoRA from Assignment 2 to test whether emergent misalignment also increases CoT unfaithfulness.

Evaluation

1. **Primary metric.** Our primary metric is the IPHR rate: the fraction of question pairs where the model gives the same answer to both logically opposite variants.
2. **Evaluation conditions.** We compare three inference settings:
 - (1) Baseline.
 - (2) Ablation of the probe-identified bias direction.
 - (3) Ablation of a random direction.
3. **Success criterion.** A successful outcome is a meaningful reduction in IPHR under Condition (2), but not under Condition (3), while preserving question-level accuracy.
4. **Interpretation of a null result.** If ablating the probe-identified direction does not reduce IPHR, this would indicate that the probed direction is correlated with, but not causally responsible for, the model’s unfaithful reasoning. Such a result would establish a concrete limitation of representation engineering as an intervention for improving CoT faithfulness.